

Existential Forgers can
have padding schemes
can protect.

- Existential signature forgery is where an attacker produces a valid message / signature pair (m', s) , without knowing private key
- An attacker has access to Alice's public key (n, e)
- They can calculate $m' = s^e \bmod n$
- This always produces a valid pair - making forgery ~~the~~ trivial on unpadded RSA
- A scheme like PSS or PKCS#1 enforces that a valid message must follow a strict recognisable structure
- A randomly forged $m = s^e \bmod n$ is unlikely to match this structure

- Adding a hash function makes this even harder - the forged m would also need to be a valid ~~output~~ hash output (an attacker cannot control)

- This makes existential forgery computationally infeasible, because the attacker can no longer freely choose s and get a meaningful m out

In a scheme:

Signer: $m \rightarrow \text{hash} \rightarrow \text{pad} \rightarrow S$
 $\text{pad}(\text{hash}(m))^{1d} \bmod n$

Verifier: $S \rightarrow S \wedge \rightarrow \text{check padding / hash matches } m$